

Special Exam (1h30)

Exercise 1 (6.5 pts). On a (4×4) -grid, we can only move to the right or upwards.

1. How many ways are there to connect point $A(0, 0)$ to point $B(4, 4)$?
2. How many ways are there connecting A to B and pass through
 - the point $P(1, 2)$? Choosing a way at random from all ways connecting A to B , what is the percentage that this way passes through P ?
 - the point $Q(3, 3)$?
 - the points $P(1, 2)$ and $Q(3, 3)$?
 - the points $P(1, 2)$ or $Q(3, 3)$?
3. How many ways are there to connect point $A(0, 0)$ to point $B(4, 4)$ if the down line is off limit?

Exercise 2 (7 pts). A permutation of $\{1, \dots, n\}$ is **non consecutive** if $12, 23, \dots, (n-1)n$, do not occur. Let Q_n , be the number of non consecutive permutations of $\{1, \dots, n\}$.

1. Give the different configurations for $n = 1, 2, 3$ and then the values of Q_1 , Q_2 and Q_3 .
2. Give the explicit formula of Q_n , for $n \geq 1$.

Exercise 3 (6.5 pts).

1. A salesman sells at least 1 car each day for 100 consecutive days selling a total of 150 cars. Show that for each value of n with $1 \leq n < 50$, there is a period of consecutive days during which he sold a total of exactly n cars.
2. A party is attended by $n \geq 2$ people. Prove that there will always be two people who have the same number of friends at the party.